

MANAS REDDY ARUMALLA

Robotics & Control Engineer | MSc Autonomy Technologies, FAU Erlangen-Nürnberg
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SUMMARY

Control & autonomy engineer specializing in estimation and safe control for aerial and mobile robots. Published author (ICRM 2025) and DST-funded researcher, with a research-grade portfolio spanning a vision-only drone that lands on moving ship decks, a bimanual manipulation platform, and 15+ benchmarked controller designs. Comfortable across the full stack — dynamic modelling and control design in MATLAB, validation in MuJoCo/Gazebo, and deployment to ROS 2 and embedded hardware. Interested in perception-driven, provably safe control for autonomous systems.

EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg
M.Sc. Autonomy Technologies — Nürnberg, Germany

2026 – Present

Amrita Vishwa Vidyapeetham

2021 – 2025

B.Tech, Automation & Robotics — Coimbatore, India | CGPA 8.84/10.0 (First Class with Distinction)

Relevant coursework: Control Systems, Robotics & Mobile Robotics, Computer Vision, Machine Learning, Industrial Automation, Embedded Systems.

TECHNICAL SKILLS

Programming: Python, C++, MATLAB

Control & Estimation: LQR, MPC, SMC, PID, backstepping, H^∞ , adaptive & predictive control; state-space modelling; trajectory tracking (Pure Pursuit, Stanley, DWA)

Planning & Navigation: A*, RRT/RRT*, PRM, Hybrid A*, APF; RVO multi-robot avoidance; PDDL / PlanSys2; ROS 2 Nav2

ML / RL: Reinforcement Learning (PPO), genetic-algorithm optimization, computer vision (OpenCV, MediaPipe); LLM / VLA integration

Robotics Frameworks: ROS / ROS 2, Gazebo, MuJoCo, CoppeliaSim, PX4 / MAVLink

Simulation & CAD: Fusion 360 / Inventor, Adams, Simscape, Simulink

Embedded & Hardware: Arduino, Teensy, Raspberry Pi, microcontroller programming, Embedded C, sensor integration

Operating Systems: Linux, Windows

PROFESSIONAL EXPERIENCE

Robotics Engineer, Humanoid Robotics Startup (Stealth)

Feb 2026 – Mar 2026

- Built robotic-manipulation simulations in Gazebo and implemented PDDL task planning, letting task sequences be validated in simulation before committing to hardware.
- Simulated and tested a K-Scale humanoid in ROS 2 and stood up the sim-to-hardware deployment pipeline.

Research Assistant, Amrita Vishwa Vidyapeetham (under Dr. Rammohan Sriramdas)

Sep 2025 – Jan 2026

- Developed a DST-funded two-wheeled hopping robot — characterized its jump dynamics and designed the stabilizing control, work published at ICRM 2025.
- Enhanced the control architecture and navigation performance of a quadruped mobile robot.
- Authored three research papers — one published, two under review.

Robotics Intern, Anvi Robotics

Jun 2024 – Jul 2024

- Designed a water-cleaning unmanned surface vehicle (USV): component selection, CAD, and ROS simulations integrating MAVLink with PX4.
- Integrated vision-based trash detection and autonomous navigation in ROS, enabling the USV to locate and route to floating debris on its own.

Robotics Intern, Sony SSUP Program

Oct 2023 – Dec 2024

- Co-developed a four-legged mobile robot for farm assistance and cattle health monitoring under the Sony SSUP program.
- Programmed and integrated multiple sensors and control algorithms, and troubleshooted hardware to keep experiments running with minimal downtime.

Robotics Intern, 7s Technologies

Mar 2024 – Apr 2024

- Designed an AR/VR motion-simulation chair with real-time control; selected components and ran load/performance calculations to hit motion-fidelity targets.

SELECTED PROJECTS

Vision-Based Drone Landing on a Moving Platform

Python, MuJoCo, CasADi

- Built a **vision-only autonomous quadrotor** that lands on moving ground rovers and 6-DOF seakeeping ship decks from onboard sensors alone — ArUco fiducials fused in a relative-state EKF with a stabilized gimbal, no ground-truth state in the loop.
- Designed the control and safety stack: **geometric SE(3)** and nonlinear / tube-DOB MPC (CasADi), a **Hamilton–Jacobi reachability safety shield**, and higher-order control barrier functions for sense-and-avoid.
- Added reinforcement-learning (PPO / LSTM-PPO) landing policies and extended the system to a **decentralized multi-drone swarm** with Kalman-consensus estimation and a CBF-QP safety filter; backed by a 112-test suite.

OpenArm — Bimanual Manipulation Platform

Python, MuJoCo

- Built a control, planning, and learning stack for the **bimanual Enactic OpenArm v2 (7-DOF × 2)** in MuJoCo: forward/inverse kinematics, Cartesian and **compliant (admittance) control**, and RRT-Connect obstacle avoidance.
- Implemented a range of manipulation skills — pick-and-place, articulated-object handling (drawers/valves), cloth folding, collision-aware bimanual hand-over, and dynamic ball-catching via **Kalman + MPC interception** — plus a learned ACT vision policy and an RL insertion suite.

NORMA — LLM-Supervised Robot Manipulation

SmoIVLA, Claude, MCP, NormaCore

- Ran a learned VLA policy (SmoIVLA) and a reasoning LLM (Claude) **together on a real 8-DOF arm** — the LLM reasons over the camera scene and supervises the policy at runtime, taking over with corrected joint commands when it stalls or fails.
- Built LLM-driven pick-and-place (no hard-coded coordinates or IK), color sorting, and gesture-based Q&A, with safety enforced by a **custom MCP server** that gates every command so the model cannot execute an unsafe move. **Won the NormaCore track** at the HackLab Berlin AI × Robotics Hackathon.

Comparative Control Study — Two-Wheeled Inverted Pendulum

MATLAB, MuJoCo, python-control

- Derived and benchmarked **15+ controllers** (LQR, MPC, H^∞ , SMC, backstepping, NDI, LPV, adaptive, predictive) for a self-balancing Segway-type robot, with every controller **auto-tuned by genetic-algorithm optimization**.
- Added energy-based swing-up and realistic state estimation, and validated each design in a MuJoCo simulation against a MATLAB state-space model.

Swarm Autonomy — Decentralized Multi-Drone Search & Pursuit

ROS 2 Jazzy, PX4 SITL, Gazebo

- Built a **decentralized multi-UAV** system for GPS-denied urban search-and-pursuit where each drone runs the same onboard stack — **visual-inertial odometry (OpenVINS)**, ESDF occupancy mapping (nvblox), planning, and control — with no central coordinator.
- Coordinated the swarm peer-to-peer over a modeled bandwidth-limited radio, allocating containment roles with **CBBA** to corner a fleeing target; runs full-stack on ROS 2 Jazzy + PX4 SITL + Gazebo Harmonic, backed by a 77-test suite.

BallPark — Classical-vs-Learned Mobile-Autonomy Benchmark

Python

- Built a benchmark that **races classical and learned autonomy on identical tasks** — every layer (perception, SLAM/mapping, planning, control, high-level decision) has a classical and a learned implementation behind one interface, scored on a shared leaderboard from point-to-point navigation up to an N-vs-N robot-soccer capstone.
- Benchmarked A*/NMPC/MPPI/SLAM against RL, imitation, and **diffusion policies** with seeded Monte-Carlo statistics, surfacing where learning helps and where it doesn't — including a classical controller cascade that **Pareto-dominates** the best optimizer on the reliability/compute trade-off. Backed by a 104-test suite.

Additional: Steady Stride — physical self-balancing robot with onboard manipulator (Arduino/C++, PID & LQR, IMU sensor fusion); TurtleBot3 ROS 2 task planning with PlanSys2 + PDDL and Nav2; manipdyn 6-DOF UR5e control & planning lab (8 controllers, 5 planners); quadrotor control benchmarks in Python & MATLAB (PID / LQR / SE(3) / SMC / MPC, GA-tuned).

PUBLICATIONS & RESEARCH

- M. R. Arumalla, R. Sriramdas, V. R. Devarakonda, A. Vincent, “Mapping the Dynamics of Robotic Jumps in Hopping Robots,” International Conference on Robotics and Mechatronics (ICRM 2025), 2025.
- Two additional manuscripts currently under review (2026).

ACHIEVEMENTS & LEADERSHIP

- **HackLab Berlin AI × Robotics Hackathon** (2026) — won the NormaCore track (team of 2) for an LLM-supervised robot-manipulation system on an 8-DOF arm.
- **e-Yantra Robotics Competition 2022-23** (IIT Bombay) — awarded Level 3 for an autonomous pharma-delivery bot.
- **Robotics Club** — led the projects team; designed and delivered a hands-on ROS workshop.
- **NSS Volunteer** — contributed to community camps and social-outreach initiatives.